Access transparency in grammar: Mediated accessibility as a comparative concept

Brett Reynolds [*]
Humber Polytechnic & University of Toronto

Jong-Bok Kim
Kyung Hee University

22nd August 2026

Abstract

Linguists use TRANSPARENCY for cases in which a relation can access a property through material that could have hidden it. This paper defines the narrower relation, ACCESS TRANSPARENCY, as a comparative concept. An application passes four gates: an independently diagnosable bearer, a distinct mediator, evidence that the relation tracks the property, and an independently established concealing counterpart in the same constructional family. Verdicts are positive, non-instance, or indeterminate, and feature summation, whole-configuration construal, and external bearers come out as non-instances. English number-transparent quantificational nouns supply the main empirical case, with a corpus pilot giving partial prospective support and a feasibility limit; transparent free relatives supply a second illustration, positive under Kim's constructional analysis and indeterminate analysis-neutrally. Spanish and Tsez extend the protocol cross-linguistically. Projectibility is secondary: the comparative concept predicts nothing by itself, and only a language-particular profile with an observable base and an identifiable stabilizer supports projection. The result distinguishes a relational analysis that earns its keep from a label that restates the facts it describes.

Keywords: access transparency; mediated accessibility; agreement; comparative concepts; number transparency; transparent free relatives; projectibility

1 INTRODUCTION

The initial puzzle is that some grammatical relations can still access a property through a structure that appears, on the surface, to block or reverse that access. Two examples from Huddleston and Pullum's *Cambridge Grammar of the English Language* (*CGEL*, 2002) make the point visible (§18.2, p. 502).

(1) a. *A number of spots have/*has appeared.*
b. *Heaps of money has/*have been spent.*

[*]Contact: brett.reynolds@humber.ca

In the first example, the surface head *number* is singular, but agreement is plural because the *of*-complement *spots* is plural count. In the second, the surface head *heaps* is plural, but agreement is singular because the *of*-complement *money* is non-count. The construction lets the complement's number/count profile reach the verb, even though the surface head's own morphological number points the other way.

CGEL calls these NUMBER-TRANSPARENT QUANTIFICATIONAL NOUNS (QNs).[1] Core members include *lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number,* and *couple*, with *majority* and *minority* at the boundary. They head NPs whose agreement override is, on *CGEL*'s analysis, the default, unlike the optional override of ordinary collectives like *team* and *family*. The narrower relation studied here is ACCESS TRANSPARENCY: access to a property through a mediator when an independently established counterpart shows how that mediator could have concealed it. The schema is:

> Umbrella schema.
> X is ACCESS-TRANSPARENT with respect to property P for relation R when P, borne by one element within X, remains accessible to R across a distinct mediator, while an independently established counterpart in the same constructional family shows how the mediator or the configuration could instead control R.

The schema has three variables (X, P, R), two participant roles that fill out X (bearer and mediator), and two controls (the diagnostic through which R's behaviour is observed, and the concealing counterpart against which the case is measured). Keeping those apart is what stops a case study from drifting. Applied to (1a):

Configuration (X)	*a number of spots*: a number-transparent quantificational noun plus its *of*-complement
Property (P)	plural count profile
Bearer	*spots*, the *of*-complement
Mediator	*number*, the morphologically singular head
Relation (R)	subject–verb agreement
Diagnostic	the number realized on *have*
Concealing counterpart	head-driven agreement: *the number of spots has appeared*

Three distinctions keep the application disciplined. The property is not its bearer: P is the plural count profile, not the word *spots*. The relation is not the diagnostic that detects it: agreement is the relation, and verb morphology is how we see it. The counterpart is a constructional alternative established without appealing to the access effect under assessment.

The application proceeds through four gates:

1. *Independent bearer*: P is diagnosable on an element inside X without using R.

[1] QUANTIFICATIONAL NOUN is *CGEL*'s label, and the paper keeps it. In the pseudo-partitive literature the same nouns are QUANTIFIER NOUNS, one of five sub-types alongside measure, container, part, and collective nouns (Van Eynde & Kim, 2023, p. 271), following Keizer (2007, p. 109). Van Eynde and Kim print "collection" for this last type; the paper standardizes on "collective", following Kim (2026a).

2. *Distinct mediator*: the bearer and mediator are distinct expressions or independently motivated structural positions.
3. *Tracking*: the observed relation demonstrably tracks P, rather than a newly constructed property of X or of the mediator.
4. *Independent counterpart*: the constructional family is individuated without reference to access transparency, and it contains an attested counterpart in which R tracks the mediator or X instead. The counterpart may be a strong default or a weaker, dispreferred option, but its evidential strength should be stated.

Coordination fails the second gate: in *Bob and Jane are happy*, plural agreement sums the conjuncts' features, and no mediator stands between a bearer and the relation. Deferred reference fails the first: the relevant referent lies outside the subject expression. A case that passes all four gates receives a positive relational verdict. A case that fails a gate receives a non-instance verdict when the failure is established; where the constructional analysis or counterpart remains unsettled, the verdict is indeterminate. The protocol is an applicability test, not a falsification condition. Profile predictions can be falsified only after a positive application has been established.

This non-instance requirement is the paper's distinctive contribution. It keeps mediated accessibility from collapsing into a relabeling of familiar facts, while the three-way verdict records when an analysis hasn't yet earned a binary answer. Everyday senses of *transparent prose* or *transparent intentions* remain outside the schema because its slots have no values to bind.

Passing the applicability test is necessary, not sufficient, for projection. A label can identify a genuine access relation and still add no predictive content beyond the facts that motivated it. The secondary question is whether a positive case supports PROJECTIBILITY (what observing some features licenses us to predict about others; Goodman, 1955). A PROJECTIBILITY PROFILE has an observable base, a set of predictions that base licenses, and a STABILIZER that helps maintain the covariance. Boyd (1991) supplies one maintenance-based account of stability. The QN class shows the shape concretely. Its determiner frames and complement restrictions support a candidate profile beyond the agreement contrast, but this paper directly audits only selected behaviours. The label ACCESS TRANSPARENCY itself projects nothing; any projection belongs to a language-particular profile, and projectibility is field-relative: a profile supports induction for a field's purposes, not for every inquiry at once.

The paper asks two linked questions: which configurations satisfy the access-transparent relation, and which positive cases support projection beyond the facts that motivated the label? The primary field is syntax, where the targets include agreement, syntactic-category licensing, complement selection, and behaviour under complement ellipsis. What this buys descriptive and theoretical grammar, and typology, is a way to distinguish an access relation from the profile-level question of what it predicts. The philosophy of science is instrumental: Goodman's projectibility, Slater's stable property clusters, and Khalidi's causal networks enter only where they change a grammatical assessment. Readers interested in the primary demonstrations can focus on §2 and §4; §5 supplies the shorter secondary profile assessment.

Our aim is constructive rather than exhaustive. The paper leaves aside most uses of TRANSPARENCY in linguistics: extraction, islands, barriers, and phases; phonological and orthographic transparency; everyday talk of *transparent explanation*; and referential or *de re*/*de dicto* transparency, where the mediator is an intensional context rather than a form with internal structure, so the schema's *through* X clause has nothing to bind to.

Mediated accessibility is more specific than clarity and broader than compositionality. Clarity is rhetorical: it doesn't single out a mediator or relation. Compositionality is the limiting case of form–meaning transparency, where a mediator's parts predict the whole. Number transparency is different: a QN's morphological number doesn't by itself predict the verb's number, and under the construction it doesn't have to.

Section 2 takes up morphosyntactic transparency, with number-transparent QNs as the main case and transparent free relatives as a second, conditional illustration. Section 3 briefly marks form–meaning accessibility (constructional compositionality (Goldberg, 2006), rated compound transparency (Bell & Schäfer, 2016), and time-sensitive lexical processing (Heyer & Kornishova, 2018)) as a boundary application where R shifts from agreement to interpretation or processing. Section 4 applies Haspelmath's (2010) comparative-concept discipline to the schema's cross-linguistic ambitions, with Spanish and Tsez as controlled cases and the fuller topology survey in Appendix B. Section 5 gives the secondary profile assessment, using Slater's stable property cluster (SPC) framework (Slater, 2015) and Khalidi's causal-network account (Khalidi, 2018) only where they sharpen the evidential verdict.

2 Morphosyntactic transparency

Morphosyntactic transparency is the case where X is a syntactic construction (or a lexical class within one) and R is a grammatical relation sensitive to a property borne inside X but not by its head. Feature matching between a non-head property and a co-occurring exponent is the paradigm, and agreement is the paradigm of that. Selection, category licensing, binding, and dependency formation are possible targets, as §2.5 shows for transparent free relatives.[2] Subject-verb agreement is the textbook instance. The concrete contrasts are simple. In (1a), agreement tracks *spots* through singular *number*; in (14a), agreement tracks *trousers* through singular *what*.

This section carries the paper's primary field-relative claim. The projected targets are syntactic and morphosyntactic: agreement, syntactic-category licensing, complement selection, behaviour under complement ellipsis, and distributional restrictions inside NP and fused-relative constructions. Nothing in the syntactic result depends on phrase-structure representation as such: the profiles require only observable relations among a mediator, an internal property, and the grammatical relation that tracks it, whether those relations are represented in constituency or dependency terms. The section contains two grammatical cases, but only the QN section carries a prospective candidate-member test; the TFR result remains a conditional illustration based on Kim's constructional analysis and pilot corpus evidence.

[2] Extraction transparency (whether an island or barrier is opaque or transparent to long-distance dependency formation) is a morphosyntactic-accessibility relation under the schema (with R = dependency formation), and the associated question remains active. Transformational accounts posit empty elements (traces, copies) that mediate the connection between a fronted element and its base position; non-transformational dependency accounts (da Cunha & Gibson, 2025; Gibson, 2025) argue that no such mediating empty elements are needed and that direct head-to-dependent connections better fit acceptability and processing data. Both sides make schema-level claims about whether X contains invisible mediating structure. The centre of the argument here is the agreement-style cases.

2.1 Number-transparent quantificational nouns

The paradigm again is *CGEL*'s two-direction contrast in (1): plural agreement despite singular *number*, and singular agreement despite plural *heaps*. Schema-wise (under the umbrella schema), X is the number-transparent quantificational noun + *of*-complement construction; P is the number/count profile of the *of*-complement; R is subject-verb agreement.

CGEL's "simple agreement rule" is head-driven: "the verb agrees with a subject with the form of an NP whose person–number classification derives from its head noun" (§18.1, p. 499), as in (2).

(2) a. *The nurse wants to see him.*
    b. *The nurses want to see him.*

Number-transparent NPs override the rule. *CGEL* describes the override as obligatory in the central cases; rare singular-verb attestations with *number* are noted in footnote 73 to §18.2 (p. 502) and analyzed there as hypercorrections rather than as variant grammar. The contrast with the optional override of collective nouns (§2.2) is the operative point. The construction's job is to let the *of*-complement's number/count profile reach agreement past the head's morphological number, which would otherwise have determined it.

The core inventory is small and well documented. *CGEL* identifies the principal members in example [57] (§3.3, p. 350): *lot, plenty* (singular forms; *lot* most often with *a*-determiner, *plenty* typically without a determiner); *lots, bags, heaps, loads, oodles, stacks* (plural forms, typically without a determiner); *remainder, rest* (singular forms typically with *the*-determiner); *number, couple*$_2$ (singular forms typically with *a*-determiner, taking only plural *of*-complements, partitive or pseudo-partitive). Of these, "the clear cases of number-transparent singular nouns are *lot, number*, and *couple*$_2$" (p. 503); *majority* and *minority* are borderline: plural override is "surely obligatory" in examples like *The majority of her friends are Irish*, but singular agreement is occasionally attested (§18.2, pp. 503–504). The determiner patterns above are typical, not absolute: *the lot of it* (in a totality sense) is grammatical alongside *a lot of it*, and similar sense-conditioned variations exist for other members.

Keizer's (2007) study of English NP categorization and Van Eynde and Kim's (2023) Head-driven Phrase Structure Grammar (HPSG) analysis of pseudo-partitives are the immediate background for treating these strings as structured binominal configurations rather than as a single loose family. Within the binominal family, Kim (2026a, pp. 98–103) distinguishes three sub-types by agreement head, illustrated in (3): measure nouns (*pound*), where the first noun controls agreement; quantity nouns (*lot*, *number*), where the second noun controls it; and collective nouns (*group*, *bunch*), which allow either.

(3) a. Measure (N1 controls): *One pound of tomatoes equals 90 calories.*
    b. Quantity (N2 controls): *A lot of marbles are about to fall on the floor.*
    c. Collective (either): *A group of police cruisers is speeding across the bridge*; *a group of immigrants move in*.

The three example sentences are Kim's, from COCA (Kim, 2026a, pp. 99–102). The quantity type is *CGEL*'s number-transparent class, and the measure type gives the schema its concealing variant: the same binominal string with agreement held by the first noun (§2.3).

Class membership predicts a cluster of behaviours beyond the agreement override. The relevant contrast is the standard PARTITIVE / PSEUDO-PARTITIVE distinction: partitive *of*-complements require a definite complement that picks out a subset of a definite set (*of the money*, *of those books*, *of these soldiers*), while pseudo-partitive *of*-complements express a quantity, measure, or collection relation with a bare or otherwise non-definite complement in the QN frame (*of money*, *of soldiers*, *of some paintings*) (Kim & Michaelis, 2020; Selkirk, 1977; Stickney, 2007; Van Eynde & Kim, 2023).[3]

Lot most often takes *a* as determiner with limited premodification (*A whole/huge lot of time has been wasted*; *CGEL*, §3.3, p. 350); *the lot of it* is also available in a totality sense, but other determiners are blocked (**some lot of it*). *Plenty* typically takes no determiner or modifier (*plenty of butter*, not generally *a remarkable plenty of butter*), and like the plural-only forms it strongly prefers the bare *of*-complement.[4] The plural-only forms (*lots, bags, heaps, loads, oodles, stacks*, and *piles* beyond *CGEL*'s list) show the same pattern.

(4) a. *They have oodles of money.*
b. *Oodles of the money had already been spent.*

The contrast in (4) shows the plural-only pattern: the first is the characteristic use, the second a marked one. It is not a blocked one. A wildcard sweep of the determined frame attests it for every member of the subgroup, at 465 tokens across 374 distinct complements, where a probe over four hand-picked complements had returned 13 and two zeros (Appendix A.2). *Remainder* and *rest* are a different case again; *number* and *couple*$_2$ take plural obliques.

(5) a. *the remainder of the time*
b. **the remainder of time* (partitive reading only; fine as "all time henceforth")
c. *the rest of society*
d. *a number of the protesters*
e. **a number of money*

The obliques in (5) are structured, but the lexical classification of *rest* is not settled by the corpus pattern. Under the semantic criterion adopted here, *rest* and *remainder* denote the complement subset of a presupposed whole rather than a quantity of it, so the pseudo-partitive construction, defined as one in which N1 denotes "a quantity or amount" of what N2 denotes (Van Eynde & Kim, 2023, p. 271), is unavailable. A competing analysis could treat bare *the rest of society* as pseudo-partitive by treating *rest* as a part or amount expression. The data establish the bounded-and-unique condition on the bare frame, but do not decide that lexical-semantic question; Table 1 therefore marks *rest* as unresolved while retaining the semantic analysis as the working description.

[3]The headedness issue is framework-internal. HPSG analyses treat the first noun as syntactic head in *A of the B* partitives, while *CGEL* retains the quantificational noun as head and treats agreement with the complement as override. Since the schema is framework-portable rather than neutral (§3), the paper doesn't adjudicate the headedness choice; it records that the verdict is conditional on it.

[4]A COCA pilot (May 2026, extended July 2026) supports the preference but not a rate. The bare complement dominates for *plenty* and the plural-only forms, while *rest* and *remainder* prefer the determined one on matched complements ($\approx$ 97% for *day, time, people, money*). No overall share is defensible: the two frames were sampled over different complement inventories, and frequency in any case supplies no denominator of occasions on which a partitive reading was intended. The partitive frame is attested for every member of both subgroups. Full counts and methodology in Appendix A.2; raw data in the supplementary material (see Data availability).

For the bare-complement reading, the whole must be uniquely identifiable *and* bounded, so that it has a proper part to take a remainder of. Generic collectives supply both unaided, which is why (5c) needs no determiner and is common: *the rest of society* alone has 442 COCA tokens, with *humanity* 248, *nature* 78, *mankind* 77, *creation* 42, and *Congress* 21 behind it. An unbounded complement supplies neither. Bare *of time* offers no partition, which is why (5b) is starred on the relevant reading and why the string survives only on the unrelated duration reading, "all time henceforth". That reading accounts for most of the corpus tokens of *the rest of time*, so those tokens are not evidence about the partitive/pseudo-partitive classification (Appendix A.2).

CGEL's own minimal pair for *lot* has the complement controlling agreement in both directions (p. 349).

(6) a. *A lot of work was done.*
b. *A lot of errors were made.*

The pair in (6) shows what number transparency *is* in the umbrella schema's vocabulary. X (the construction) leaves P (the number/count profile of the *of*-complement) accessible to R (agreement). What X conceals from agreement is the morphological number of the head noun *lot, plenty, heaps*, etc. The agreement-targeting feature is on the *of*-complement, not on the surface head. That's what makes X transparent with respect to P: X lets P reach the verb anyway.

The override is only visible where the head's number and the complement's diverge. (6b) shows it (singular *lot*, plural *errors*); (6a) doesn't, since head and complement both call for the singular. The discriminating cases are the crossed ones: a singular-form quantificational noun with a plural complement, or a plural-form one (*lots, heaps*) with a non-count complement. The matched cases are silent, not counterevidence. Transparency is therefore under-detected: a class can be transparent while most of its tokens look head-driven, so corpus frequency understates it.

2.2 Collective NPs as a transitional case

Singular collective nouns (*committee, jury, family, team, government*; also *clergy, intelligentsia, public*, etc.) allow optional plural override of head-driven agreement, as in *CGEL* [8] and [10] (§18.2, pp. 501–502).

(7) a. *The committee has/have not yet come to a decision.*
b. *The committee consists/*consist of two academic staff and three students.*
c. *One committee, appointed last year, has/?have not yet met.*

The override in (7) is construal-sensitive: singular agreement tracks the unit construal, plural the member construal. Plural is unavailable with predicates that apply to the whole but not individual members, and is "of questionable acceptability" if the collective takes determiner *one* or *another* (p. 502).

Both collective NPs and number-transparent quantificational NPs override the head's morphological singular at agreement, and the override is possible because plural agreement requires only non-singular construal, not the exact enumeration that singular *a*(*n*) and low cardinals demand. That is why tight determiners degrade the override that (7a) allows while plural agreement survives it: the override operates at the loose end of the count cluster, not across it. Reynolds (2026a) develops that loose/tight dissociation at length.

The two cases differ in where the non-singular construal comes from. In collectives it is lexically encoded in the head, so the override is selective: it needs a predicate that licenses member-level construal, which is why whole-unit predicates block it. In number-transparent QNs it is imported from the *of*-complement, so the override applies regardless of predicate type. That difference matters more than it first appears, because only one of them satisfies the schema as stated. The schema requires one element within X to bear P and another to conceal it. In the QN case those are distinct expressions: the *of*-complement bears the number profile, the quantificational head conceals it. In *the committee have decided* they are not. The singular morphology and the member-level plurality are two representations of one lexical head, so there is no second element doing the concealing. Collectives are therefore a near case: a competition between morphological and semantic properties within a single sign, not access across a mediator. Broadening *element* to cover representational layers would accommodate them, at the cost of the non-instance test's grip, since almost any property could then be said to conceal another representation of itself. The transitional label in this subsection's title is meant in that strict sense.

Alongside *CGEL*'s override framing, the HPSG tradition encodes the same observation as index rather than concord agreement: concord tracks morphosyntactic features internal to the NP, while index agreement can track the referential or semantic index available to the agreement target (Kim, 2004; Wechsler & Zlatić, 2003). Kim and Sells (2015, pp. 64–65) run the two levels on the cases at issue here: measure *four pounds was* is morphosyntactically plural but index-singular, while collective *this team have* is the mirror image, morphosyntactically singular with a plural index. On that encoding, collective agreement makes semantic plurality available to agreement, but it isn't access transparency in the strict sense used here, because the bearer and the proposed mediator aren't distinct elements. The index/concord split is consistent with the framework-portability of §3.

CGEL itself notes that "the division between collective and number-transparent nouns is by no means sharply drawn" (§18.2, p. 502). Some nouns straddle. *Couple* has two senses: *couple*$_1$ (a man and woman in a relationship) is a regular collective; *couple*$_2$ ('approximately two') is number-transparent. *Bunch* heads the same binominal *of*-frame as §2.1's quantity nouns but is a collective noun in Kim's (2026a, pp. 98–103) classification, the sub-type where either noun can control agreement; the straddling *CGEL* reports is what that classification predicts. *Bunch* is also the case where the profile framework does demotion work: *CGEL* claims it tilts collective for inanimate aggregates but transparent for groups of people, illustrated by the contrast in (8).

(8) a. *A bunch of flowers was presented to the teacher.*
 b. *A bunch of hooligans were seen leaving the premises.*

CGEL treats the first as collective and the second as transparent (§18.2, p. 503). A pilot COCA check (May 2026) supports the animate half, where *bunch*-subjects are consistently plural, and finds no positive support for the inanimate half: inanimate *bunch* is rare in subject position, *CGEL*'s own example is unattested, and the few tokens that surface take plural rather than the predicted singular. Corpus absence of a constructed example isn't ungrammaticality, so this doesn't touch *CGEL*'s contrast as a claim about grammar; what it shows is that the inanimate-collective pattern doesn't project as a corpus regularity. That localizes the projectibility claim rather than the grammatical one, and marking where a textbook contrast doesn't project is what demotion conditions are for. Counts in Appendix A.4.

The paradigm and the transitional case also differ in internal structure. Number-transparent NPs are class-like, with the override the default behaviour for class members. Collective NPs are gradient: the override probability depends on the head, the predicate, the determiner, the dialect register, and the construal the speaker has in mind. A theory of morphosyntactic transparency that ignored the difference between a class-based default override and a construal-driven optional one would lose the interesting structure of both cases.

2.3 Coordination as a non-instance

Coordination of singulars triggers plural agreement.

(9) *Bob and Jane are happy.*

The ordinary coordination in (9) looks like a transparency case at first (the conjuncts' number features percolate up to the coordinate construction and determine agreement), but the schema excludes it. The schema requires that X *could have* made P inaccessible. The coordinate construction does no such hiding: there's no head whose morphological number competes with the conjuncts' features for agreement.[5] The plural agreement is compositional addition (1 + 1 = 2), not mediation.

Some coordinate NPs take singular agreement.

(10) a. *Bacon and eggs is my favourite breakfast.*
b. *Wine and cheese is expected.*
c. *Fish and chips is expensive these days.*

Some of the same coordinate strings also take plural agreement.

(11) a. *Wine and cheese make the perfect party pair.*
b. *Fish and chips taste their best right after frying.*

The singular holistic cases in (10) and the plural counterparts in (11) leave the exclusion intact. In such cases agreement tracks a construal of the coordinate phrase: either a dish, package, event type, or conventional pairing, or a plural set of jointly mentioned items. The singular feature isn't a property of either conjunct that remains accessible through the coordinate construction; it's a construal of the coordinate phrase as a whole. Ordinary plural coordination and holistic singular coordination therefore differ in agreement semantics, but neither is transparent in the schema's sense. The former is compositional summation; the latter is whole-phrase unit construal.[6] *CGEL* treats coordination-and-agreement under a separate heading (§18.4), not alongside the override constructions of §18.2.

The clean distinction across §2's cases: in number-transparent QNs, complement number becomes accessible through a quantificational head; in collective plural agreement, member plurality

[5] This verdict presupposes a headless (or numberless-head) analysis: with no conjunct serving as a number-bearing head, nothing conceals the conjuncts' summed features. The picture changes under an analysis on which one conjunct is the head. Closest-conjunct agreement (Bhatt & Walkow, 2013) is the case in point: where a verb tracks the nearest conjunct, that agreement conceals the resolved features and resolved agreement reaches them past it, so the same test can return an instance. As with the language-internal concealing baseline of §4, the verdict rides on the prior syntactic analysis.

[6] Cases like *One and one is two* are an arithmetic-expression construction, not a meal/package coordination: *and* functions close to an addition operator, and the subject denotes an operation, equation, or expression. Not transparency either.

is available to agreement but is borne by the head itself, so no distinct mediator intervenes; in unit-construal coordination, the whole coordinate phrase is construed as singular and no internal property is being made transparent. Only the first is access through a mediator.

The "X could have made P inaccessible" clause works as an independent diagnostic, not an ad hoc filter. The test asks whether the construction-type admits a concealing counterpart: is there a version of the same construction-type, attested in the language, in which X's internal structure makes P inaccessible to R? For English NPs, head-driven agreement is the default and the number-transparent class is the marked exception within the same NP type. For coordinate constructions no such counterpart exists: the type admits no version in which conjunct features fail to reach agreement, so coordination has no control against which access can be measured. The diagnostic distinguishes mediator-types from compositional-summation types: access is informative only where the same mediator type has opaque or partially opaque alternatives.

One condition has to be attached, or *same construction-type* becomes adjustable after the fact. Drawn narrowly enough, no transparent construction has an opaque sibling and every case is a non-instance; drawn broadly enough, almost anything does and the filter passes everything. So the concealing control must belong to a constructional family that is independently motivated, established on criteria that don't appeal to the transparency effect the test is assessing. English head-driven agreement qualifies: it is the unmarked pattern for NP subjects on grounds that have nothing to do with quantificational nouns, which is what makes the number-transparent class a marked exception *within* a family fixed elsewhere. Tsez class IV agreement with a clausal argument qualifies on the same footing. Where no such family can be identified without invoking the effect, the test returns no verdict rather than a negative one; the typological cases in §4 show why that control has to be fixed language-internally.

The condition also locates where the diagnostic is soft. Deciding what counts as the same construction-type is itself a language-internal analysis, so the test is crisp where the family is independently motivated and loose where it isn't. That is a real limitation on the paper's central instrument, not a detail about its edges.

A second non-instance fails on a different role, and the contrast is worth having because it answers the obvious deflationary reading of the QN case. In deferred or metonymic reference, agreement can track a referent the sentence never names.

(12) a. *The hash browns at table nine are/*is getting cold.*
b. *The hash browns at table nine is/*are getting angry.*

Kim's (2004, p. 1108) pair, which he credits to Nunberg, turns on which entity the subject is used to pick out: the food in the first, the customer who ordered it in the second. Warren (2006, p. 16) states the generalization, that metonymic subjects "need not agree as to number with their predicates" where metaphorical ones consistently do, and adds that a metonymic pronoun "sometimes agrees with the explicit and sometimes with the implicit element of the expression", the choice settled by context.

Filled into the frame of §1, the *bearer* cell is empty. The singular in (12b) is sourced from the designated customer, who is expressed nowhere in the subject NP, so there is no element within X bearing P and nothing for the relation to reach *through*. Coordination fails the test for want of a mediator; this fails for want of a bearer. Both are outside the schema, for different reasons.

Number transparency can be read as construal-tracking: the verb agrees with how the speaker conceives the subject, rather than with any constituent. Reference transfer shows what construal-tracking looks like on its own, and it looks different. It's context-dependent and reversible on one string, it isn't tied to a lexical class, and on Warren's account the pronoun may go either way. The QN override is none of these: near-categorical for core members, tied to a small conventional class, and construction-linked, as Appendix A.5's *a number of* versus *the number of* control shows on a single lexeme. Same surface phenomenon, different mechanism, and the frame's cells say which.

2.4 Number-transparent NPs as a projectibility profile

Section 1 set the secondary question: does identifying a positive access relation let us predict behaviour? For number-transparent quantificational nouns the profile hypothesis can be tested in two task settings: description of known members and class discovery for candidates. (The framework returns negatives too: the inanimate half of the *bunch* contrast fails to project, §2.2, and coordination never qualifies, §2.3.)

For known *CGEL* members (*lot, plenty, lots, bags, heaps, loads, oodles, stacks, remainder, rest, number, couple*$_2$), lexical identity is sufficient input for description: knowing a noun is on this list licenses comparison with the rest of the profile. The non-trivial projective test is candidate discovery, where the input criteria cannot simply repeat the targets.

For candidate new members, class discovery uses two diagnostics independently observable from any of the projected behaviours: quantificational semantics (an indefinite quantity rather than a precise count or kind) and high-frequency occurrence as head of an *of*-complement construction. A candidate that satisfies both is then tested against the projected cluster:

- near-obligatory agreement override in the central cases, singular- or plural-direction depending on subgroup;[7]
- characteristic determiner patterns by subgroup (*lot* most often with *a*, also with *the* in totality uses; *plenty* typically without a determiner; the plural-only forms typically without a determiner; *remainder, rest* typically with *the*; *number, couple*$_2$ typically with *a*);
- partitive vs. pseudo-partitive *of*-complement availability;
- behaviour under complement ellipsis: the *of*-complement can be omitted while remaining understood, and it continues to determine the NP's number (*A lot was spent on travel* vs *A lot were arrested*; *CGEL* [56], p. 349);[8]
- count/non-count percolation of the *of*-complement to the whole NP (*lot* "allows the number of the oblique to percolate up to determine the number of the whole NP"; *CGEL*, p. 349), with consequences for selection by predicates that are count- or non-count-sensitive;
- restricted premodification (*a whole/huge lot* but not *a remarkable plenty*).

The class's sub-group structure makes the cluster's predictions concrete. Table 1 displays the per-noun pattern for the central diagnostics. The entries are *CGEL*-coded except in the two *of*-

[7] A pilot COCA check found every fully audited discriminating cell in the predicted direction. Counter-direction cells received more extensive manual filtering than predicted-direction cells, so the counts are attestational corroboration rather than estimates of grammatical rates. *A lot of money* required parse-shift filtering, and the large *a lot of people is/was* cell remains incompletely scrubbed. Full counts and methodology: Appendix A.3.

[8] These bare uses are not *CGEL*'s FUSED-HEAD construction, where the head fuses with a determiner or modifier (*Did you buy* [*some*] *yesterday?*; §9.1, p. 410). *CGEL* analyzes bare QN NPs as ordinary heads with an elliptical complement (pp. 412, 503). The fusion concept returns, differently applied, in the fused relatives of §2.5.

complement columns, where the COCA pilot has revised them: the determined frame is attested for every plural-only member, and *rest* and *remainder* admit a bare complement when it denotes uniquely. The semantic classification of *rest* remains open, and a fuller per-noun corpus audit is the within-domain check still needed for the profile claim.

Table 1: Number-transparent quantificational nouns: behavioural matrix (*CGEL*-based pilot).

Sub-group / Noun	Determiner	Partitive *of* N	Pseudo-partitive *of* N	Override direction
Singular, *a*-determined				
lot	*a*; also *the* (totality)	✓ (*a lot of the money*)	✓ (*a lot of money*)	sg → pl with plural complement
number	*a*	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement
couple$_2$	*a*	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement
Singular, usually without a determiner				
plenty	usually without a determiner	not typical	✓ (*plenty of money*)	sg → pl with plural complement
Plural, usually without a determiner				
lots, bags, heaps, loads, oodles, stacks	usually without a determiner	✓, dispreferred; attested for every member (§2.1)	✓	pl → sg with non-count complement
Singular, *the*-determined				
remainder	*the*	✓ (*the remainder of the time*); bare whole only if bounded and unique (**the remainder of time*)	n/a: not a quantity noun	sg/pl (complement-controlled)
rest	*the*	✓ (*the rest of the time*); bare whole only if bounded and unique (*the rest of society*)	? (classification disputed)	sg/pl (complement-controlled)
Boundary				

Sub-group / Noun	Determiner	Partitive *of* N	Pseudo-partitive *of* N	Override direction
majority, minority	*a*	✓	?	variable
bunch	*a*	✓	✓	variable

One sentence per central sub-group shows the override directions the table codes.

(13) a. *A number of spots have appeared.*
b. *Plenty of people were stranded.*
c. *Heaps of money has been spent.*
d. *The rest of the money was returned.*

(13a) and (13c) repeat (1): a singular *a*-determined noun driven plural by its complement, a plural-only noun driven singular. (13b) shows determinerless *plenty* with the plural override; (13d) shows the *the*-determined sub-group, where direction is complement-controlled (plural instead with *the rest of the people*). The directions in all four match the COCA pilot's counts in Table 2.

CGEL treats override as obligatory in the central cases; the pilot corroborates the direction in selected cells but does not audit every class member or estimate rates. Complement-ellipsis behaviour and count/non-count percolation remain *CGEL*-based predictions rather than per-noun corpus results. Premodification is restricted across the class, though the restriction looks like one on modifier type rather than on premodification as such: degree and size modifiers are available (*a whole/huge lot*; *huge*, *enormous*, and *vast piles of money*), while evaluative ones are marginal (*a remarkable plenty*), and *plenty* takes minimal modification generally.

Each sub-group covaries: knowing a candidate noun's determiner pattern and partitive availability narrows the sub-group, which then licenses the override-direction prediction (and vice versa). One qualification on the partitive diagnostic. For *remainder*, the absence of a quantity reading follows from what kind of noun it is; for *rest*, the lexical-semantic classification remains unresolved. The diagnostic therefore carries less weight than the determiner and override behaviours (Appendix A.2). The boundary cases have weaker covariance, which is why §2.2's gradient framing applies to them.

Table 2 displays the raw agreement-direction counts. The discriminating cells are the informative ones, where the head's number and the complement's diverge. The matched cells confirm the expected direction but cannot distinguish the override from head-driven agreement. Counter-direction cells received more extensive manual checking than predicted-direction cells, so the table reports an audit trail rather than estimated rates. The large *a lot of people* cell remains asymmetric, and the smallest cells remain feasibility checks. Referential *the number of* is a negative control: the same noun agrees head-driven under *the* and overrides only under quantificational *a* (Appendix A.5), so the contrast belongs to the construction, not the lexeme.

This is the prospective design for a new-member case: quantificational semantics and *of*-complement-head distribution are inputs, while the cluster of agreement, determiner, partitive, ellipsis, count/non-count, and premodification behaviours supplies the targets. Boundary cases like *majority* and *minority* (§2.2) and the dual-sense *couple*$_1$ (collective) versus *couple*$_2$ (number-transparent) show graded membership. The design distinguishes prediction from redescription, but the present evidence supports only selected targets.

Table 2: Raw agreement-direction counts in the COCA pilot. Counts are after the filtering described in Appendix A.3; the audit-status column records the evidential limitation.

Subject	Predicted	Plural	Singular	Audit status
a lot of people	plural	4,195	85	Large counter-direction cell not symmetrically scrubbed
a lot of money	singular	1	90	Counter-direction cell checked; parse-shift filtering
a number of people	plural	100	0	False positives checked
lots of people	plural	348	0	False positives/non-standard uses checked
lots of money	singular	0	24	Small cell; false positive checked
plenty of people	plural	79	0	Small cell
plenty of money	singular	0	6	Small cell
the rest of the people	plural	19	0	Small cell
the rest of the money	singular	0	19	Small cell

The more demanding test asks what the profile predicts beyond the in-grammar cluster. Diachronic entry and cross-linguistic analogues are plausible targets, but neither is tested here. One candidate-member test has been run. *Piles* was admitted on two criteria only, quantity meaning and frequent *of*-complement distribution, neither of which is among the predicted behaviours. The determiner-frame prediction receives partial support: *a piles of* returns no COCA hits, as the plural-only subgroup requires, while the determined frame *the piles of* N selects the literal reading in the available sample. This result separates the two senses of *piles*, but it does not validate the full profile.

A follow-up sweep suggests that the separation extends across the plural-only subgroup. The determined frame recruits the literal homonym wherever a member has one: *stacks of the* * is dominated by library stacks and smokestacks, *heaps of the* * by literal and colliery heaps, and *bags* and *loads* by containers and cargo. *Oodles*, the one member with no literal homonym, returns quantificational hits exclusively. This is a useful subgroup hypothesis, not a completed projection test.

The agreement prediction couldn't be tested at all. COCA has 91 tokens of *piles of money* and one followed by a finite form of *be*; adding COHA's two centuries brings the discriminating non-count cell to three tokens, one singular and two plural. Nothing there supports the override and three tokens cannot weigh against it either. A container-noun control (*sacks/boxes/crates of money*) returns nothing at all, and the left contexts of the 91 tokens locate the problem precisely: 41 have *piles of money* as the object of a verb (*make, spend, earn, amass, inherit*) and 13 as the complement of a preposition, so at least 59% sit in complement position, while the single token preceded by finite *be* is copular or existential, with agreement running from a different subject. Quantity expressions of this shape occur as complements, not as subjects of finite *be*. The same sparsity blocked the inanimate *bunch* contrast in §2.2.

So the override prediction, which is the profile's headline, is corpus-testable only for members that appear in subject position, and candidate-member discovery for this class will need acceptability judgments rather than frequency. The *piles* result is therefore partial prospective support plus a feasibility failure. The protocol, full returns, and corrections the wildcard queries forced on Appendix A.2 are in the supplementary material (see Data availability).

The cluster makes number-transparent quantificational nouns the strongest candidate PROJECTIBILITY PROFILE examined here: an observable base supports projection to a structured set of targets, with conventional lexical inheritance as the proposed stabilizer. The profile is a candidate for Slater's (2015) stability-plus-projection diagnostic, but the within-domain check beyond selected agreement and determiner behaviour remains open. A profile prediction would fail if a noun satisfying the input diagnostics systematically lacked the target cluster. The status of *rest* is an explicit analytical fork: it remains an access-transparency case under either treatment, but if it is a part/subset expression rather than a quantity noun, it should not count as input to the quantity-QN candidate-discovery generalization. Whether the profile earns a local-kind label is a downstream question for §5.

2.5 Transparent free relatives

A second morphosyntactic-transparency case fits the schema. Transparent free relatives (TFRs) are a sub-type of fused-relative construction (*CGEL*, ch. 12, §6, pp. 1068ff.) in which the matrix accesses properties of an embedded predicate (the NUCLEUS) through *what*: *what they call ecstatic*, *what we regard as essential*, *what seems to be a tourist*. FUSED is *CGEL*'s term for one word jointly realizing two functions: in a fused relative, "the head of an NP fuses with the relativised element in a relative clause" (p. 1073), so *what* in *what she wrote* is at once the NP's head and the relativized element. This paper uses TFRs only as a second profile case. The analysis followed here is Kim's construction-grammar one, developed across Kim (2011), Kim (2017), Kim (2026a, ch. 6.8), and its revision in Kim (2026b, §5), with the corpus material in Kim and Reynolds (2026). The verdict below is conditional on that analysis: rival treatments locate the mediator elsewhere, and where the mediator sits is exactly what the schema needs fixed, so a reader who prefers Wilder's (1999) parenthetical-with-backward-deletion account, van Riemsdijk's (2006) graft analysis, or Grosu's (2003) unified treatment should expect the slots to be filled differently: each locates the mediator, and so the access path, somewhere else. The verdict is therefore analysis-relative: positive under Kim's constructional analysis, indeterminate on an analysis-neutral reading, since the competing treatments aren't adjudicated here.

(14) a. *What appear to be trousers are on the chair.*
b. *What they call essential workers are exhausted.*
c. *He was utterly penniless and what they call dimwitted.*

TFRs provide a second, conditional grammatical illustration, but the evidence is deliberately bounded. They support two relations at different levels of the construction. One is agreement: matrix agreement accesses the nucleus's number profile through singular *what*. The other is matrix selection, whose accessible property is the nucleus's syntactic category; CATEGORY FLEXIBILITY is the outcome, that a non-NP nucleus can satisfy the matrix's requirement, as in (14c), where the nucleus is an AP. Predicative position admits both NPs and APs, so that example shows the nucleus's category rather than establishing a category-discriminating environment. Kim (2026b, pp. 36–37) treats the second as the marked sub-case of a wider SELECTION TRANSPARENCY, on which the matrix's selectional requirement is satisfied by the nucleus generally, NP nuclei included, as when a preposition selects the nucleus through the *what*-clause (*he speaks in what linguists call a Northern dialect*).

The concealing counterpart for parent-level agreement is the ordinary fused relative, where *what* itself controls agreement. That control is family-level rather than minimally matched: an ordinary fused relative like *what she wrote* lacks a nucleus, so it can't hold the concealed property constant while varying only the access. It establishes that the fused-relative family admits mediator-controlled agreement, not that a matched variant conceals this nucleus's number. The fourth gate requires the counterpart's evidential strength to be stated, and it enters here at that weaker, family-level grade.

The pilot reported with the joint analysis supplies a bounded check on the category contrast. In a full coding of 485 archived *call* hits, 185 were genuine AP-nucleus TFRs, 144 of them externally predicative, with one token coordinated into attributive position. Exhaustive checks of the candidate sets for right-edge non-NP nuclei with *seem* or *appear* (721 tokens across eight queries, run or re-verified against COCA in August 2026) found no adjectival or adverbial case and a single PP-nucleus token in supplement position; the working paper reports the query details. This is distributional evidence, not a completed acceptability study; the relevant category-discriminating environments and judgments remain to be added to the joint paper.

The observable base is constructional sub-type plus verb-class membership, and the examples instantiate both classes Kim (2026b, p. 40) distinguishes: the EVIDENTIAL type (*seem to be, appear to be, happen to be*; 14a), which takes NP nuclei only in the current analysis, and the ATTRIBUTIONAL type (*call, consider, describe as, regard as, take to be*; 14b, c), which licenses AP, AdvP, PP, and *-ing* nuclei as well. On the inheritance hierarchy Kim (2026b, p. 46) sets out, the parent licenses agreement, selection, binding, and coordination transparency across both classes, and the attributional daughter adds category flexibility. "Coordination transparency" there is Kim's name for a TFR diagnostic, that the construction conjoins as its nucleus's category; it doesn't reverse §2.3's verdict on ordinary coordinate NPs, which concerns a different configuration. The pilot therefore supports a provisional parent/daughter contrast, with constructional inheritance as the proposed stabilizer. The full verb-class inventory, AP/PP stratification, and rival analyses remain part of the joint paper.

2.6 Bridge

The cases in Section 2 share a relation: morphosyntactic matching mediated by a construction or lexical class. Section 3 marks form–meaning applications at the boundary; the cross-linguistic reach of the schema, including the controlled Spanish and Tsez cases and the topology survey, is Section 4's topic.

3 Boundary applications beyond grammar

The schema can also describe form–meaning access when R is interpretation, learning, or processing rather than agreement. Constructional compositionality (Goldberg, 2006), rated transparency in noun–noun compounds (Bell & Schäfer, 2016), and masked morphological priming (Heyer & Kornishova, 2018) supply possible applications. In each case, the analyst would need to identify a bearer, a distinct mediator, a relation, and a constructional counterpart that could conceal the relevant property.

These examples mark scope rather than deliver additional worked profiles. Their relations, items, diagnostics, and proposed stabilizers differ, and the available studies don't establish a common within-items cluster. Constructional interpretation, compound ratings, and morphological priming belong to a research programme outside the evidential spine of this paper. The examples show why the schema keeps R explicit: a construction may be accessible for one relation and opaque for another. They don't warrant a cross-domain projectibility claim here.

4 Typological transparency

This section is methodological. The previous sections worked through cases in English; §3 marks form–meaning applications without using them as evidence for a cross-domain profile. The question here is how to move from language-particular cases to a cross-linguistic claim. Haspelmath's (2010) descriptive-versus-comparative-concept distinction is the right discipline.

4.1 Haspelmath's distinction

Haspelmath (2010) argues that crosslinguistic comparison shouldn't assume universally available descriptive categories[9] like ADJECTIVE, PASSIVE, ACCUSATIVE, or SUBJECT. Each language has its own descriptive categories, fixed by the language-particular criteria that make them work in that language (Haspelmath, 2010, p. 664). Comparative concepts, by contrast, are analyst-created tools for comparison; they aren't part of any language system and can only be more or less well suited to the comparative task, rather than right or wrong (Haspelmath, 2010, p. 665). They have to be built from universally applicable building blocks: conceptual-semantic notions, general formal notions, and other comparative concepts.

One pair makes the relation concrete. Haspelmath's comparative concept "dative case" is defined semantically, as a marker among whose functions is coding the recipient of a physical-transfer verb (Haspelmath, 2010, p. 666). Language-particular categories then match it or not independently of their traditional names: the Finnish Allative matches, while the Nivkh Dative-Accusative doesn't, despite the name, because it codes a causee rather than a recipient (Haspelmath, 2010, pp. 665–666). Categorial particularism, Haspelmath's preferred position, holds that descriptive categories are language-particular and that comparison runs through analyst-defined comparative concepts standing in many-to-many relations with them. The position it rejects, categorial universalism, assumes a substantial set of crosslinguistic descriptive categories from which languages select (Haspelmath, 2010, p. 663); Newmeyer (2007) is its canonical statement, taken up in §4.5.

The methodological consequence is straightforward. A typological generalization isn't a claim about a descriptive category instantiated in every language. It's a claim about how analyst-defined comparative concepts are realized in language-particular ways. That nominative case-marking is widespread doesn't entail that "nominative" picks out the same descriptive category in any two languages.

4.2 The umbrella schema as a comparative concept

The umbrella schema is a comparative concept, not a descriptive category. It's built from the materials Haspelmath (2010, p. 665) admits: *property* and *relation* are conceptual-semantic primitives, *form* is a general formal notion. Asking the schema's question of a language isn't claiming that the language has a category corresponding to TRANSPARENCY.

Accessible through is the contested slot, since accessibility has discipline-specific senses (discourse-pragmatic salience, psycholinguistic retrievability, formal-semantic substitution-licensing). The schema's sense is more abstract: P is accessible to R through X when R tracks P across X rather than tracking a property of X itself. The case studies operationalize that differently, by agreement

[9] Throughout, *category* is used in Haspelmath's *descriptive category* sense: a language-particular grammatical grouping fixed by a language's own distributional criteria, and neutral on whether that grouping is projectible. This differs from the sense used elsewhere in this research programme, on which a *category* just is a projectible kind (Reynolds, 2026b); that kind-level notion is carried here by the *projectibility profile* (§5.1), not by *category*.

targeting, constructional interpretability, rated transparency, or priming magnitude, but the comparative concept lives above the operationalizations.

The non-instance test (§2.3) is what keeps the concept honest, because its counterpart is fixed language-internally. The test asks whether the language under study attests a same-type variant of X whose structure makes P inaccessible to R. For English number-transparent NPs that counterpart is head-driven agreement, and the transparent class is the marked exception against it; the pattern isn't exported. The procedure is identical in any language: identify X, P, and R by that language's own distributional criteria, then look for the concealing variant. Where it exists, the transparent case is informative; where the construction-type admits none, as with English coordination, the schema returns a non-instance. So *accessible through* imports no English-particular grammar. It inherits whatever counterpart the local construction-type supplies.

So the bindings are language-particular while the question isn't. In English, X is the quantificational noun plus *of*-complement construction, a *CGEL* descriptive category; P is the complement's number/count profile; R is subject-verb agreement. In the worked cases below, Spanish partitive subjects and Tsez complement clauses fill the slots by different language-particular criteria. That's what makes the typological question tractable: not "do all languages have number-transparent constructions?", which would look for English-style NPs everywhere, but "which configurations in this language let a property remain accessible to a relation that the surface form might have hidden?" The answers are expected to differ, and the English, Spanish, and Tsez cases aren't instances of one descriptive category.

Two limits on what that buys. The claim at the comparative-concept level is structural, that the concept lives above the descriptive category whatever the language, rather than a demonstrated comparative result: the extensions on hand are the Spanish and Tsez cases below. The form–meaning family (§3) isn't itself a comparative concept; the schema is, and the family is a set of language-particular phenomena and measurements compared through it.

4.3 Cross-linguistic cases and topologies

Two worked languages let the question be asked outside English, one of them outside the agreement type English uses. Spanish has an analogous agreement-access pattern. In a partitive subject like *la mayoría de los manifestantes* ('the majority of the demonstrators'), the finite verb may agree in the plural with the complement or in the singular with the quantificational noun *mayoría*, as in (15).

(15) a. La mayoría de los manifestantes gritaban.
the majority of the demonstrators shout.IPFV.3PL
'The majority of the demonstrators were shouting.'

b. La mayoría de los manifestantes gritaba.
the majority of the demonstrators shout.IPFV.3SG
'The majority of the demonstrators was shouting.'

The plural (15a) is the more frequent option and the only one available under a plural predicative (Real Academia Española, n.d.). Both variants are attested on the one subject, much as English *majority* and *minority*, *mayoría*'s boundary cognates, take occasional singular agreement alongside the dominant plural override (§2.2). Spanish has its own literature on the alternation, which treats it as CONCORDANCIA AD SENSUM and reaches two conclusions that matter here. First, both options are genuine agreement rather than agreement displaced by government, so the plural isn't a species of SILEPSIS (San Julián Solana, 2018a). Second, the alternation isn't free across syntactic contexts, and singular agreement with the quantifier noun is compatible with a distributive reading of the clause (San Julián Solana, 2018a), which blocks the inference that notional plurality forces the override. The partitive and pseudo-partitive structures the alternation runs on are themselves heterogeneous in Spanish (San Julián Solana, 2018b), as they are in English (§2.1).

What travels to Spanish is the access pattern, not the English cluster: no Spanish determiner, complementation, ellipsis, or premodification profile has been shown here. And the concealment control is weaker than the English one, since the singular option with *mayoría* is dispreferred rather than blocked.

Tsez (Nakh-Daghestanian; head-final and ergative; about seven thousand speakers) gives a case in a different marking type. Tsez verbs prefix-agree in noun class with their absolutive argument, so the agreement is marked on the verbal head. A clausal argument is class IV, the class that includes clauses, so a matrix verb's default is class IV agreement with the complement clause as a unit; but when the embedded absolutive is a topic, the matrix verb can instead agree with it (Potsdam & Polinsky, 1999, p. 436).

(16) a. enir [užā magalu b-āc'ru-łi] r-iyxo
mother [boy-ERG bread.III.ABS III-ate-NMLZ] IV-knows
'The mother knows the boy ate the bread.' (default: the matrix verb is class IV, agreeing with the clause)

b. enir [užā magalu b-āc'ru-łi] b-iyxo
mother [boy-ERG bread.III.ABS III-ate-NMLZ] III-knows
'The mother knows the boy ate the bread.' (long-distance agreement: the matrix verb is class III, agreeing with embedded *magalu* 'bread')

In the schema's terms, X is the embedded clause, P is the noun class of the embedded absolutive, and R is matrix verb agreement. The concealing variant, and here genuinely the default, is the class IV agreement in (16a), attested in the same language and the same sentence, and (16b) is the transparent override. The counterpart isn't "the nearest head's own feature", as with English number-transparent nouns, but "default class IV for a clausal argument": a different default in a different language type. That's what the comparative concept predicts, the question constant and the language-particular answer not. Whether the override is analysed as covert movement or as long-distance agreement is a framework-internal question the schema brackets, as with extraction in §2; the fuller analysis is in Polinsky and Potsdam (2001).

Spanish and Tsez are both outward endpoint cases: a property associated with material inside X reaches an external agreement relation. The same schema maps several other accessibility topologies, provided each remains subject to the §2.3 counterpart test: identify X, P, and R, then ask what same-language pattern would make the proposed access disappear. Appendix B sets out four of them with their applicability and status conditions (null-source subpart access in Aleut, inward dependent access in Lardil and Kayardild, path-distributed access in Chamorro, and the feature-relay boundary case in Kutenai and Algonquian obviation), along with a targeted check of head-marked systems.

Two results from that survey bear on the argument here. Aleut is a qualified positive: the controller of R can be recoverable as a subpart of a non-subject NP without being overt, though Merchant's analysis treats the unpronounced possessor as syntactically present pro, which keeps the case distinct from overt intact-NP transparency (Merchant, 2011, p. 195). And the head-marked candidates cluster around possessors, most of them analysed as externalized, by possessor raising, applicative mediation, or external possession, rather than as agreement with a dependent that stays NP-internal (Deal, 2017; Ritchie, 2016). So the diagnostic point is narrow: overt sensitivity to a possessor isn't enough, and the decisive question is whether an embedded dependent remains NP-internal while controlling clause-level morphology.

4.4 Mapping transparency and access transparency

There's an established typological programme under the name TRANSPARENCY, and it classifies the paper's paradigm cases as non-transparent. Setting the two side by side is the clearest available demonstration that TRANSPARENCY isn't one projectible kind.

Leufkens defines transparency as "a one-to-one relation between meaning and form", treating every structure that violates such a relation as non-transparent (Leufkens, 2015, p. 2), and ranks 22 languages by how many non-transparent features their grammars show; the framework is extended to a larger sample and an implicational hierarchy in Hengeveld and Leufkens (2018). Opacity divides into four categories according to how the one-to-one relation fails (Leufkens, 2015, 16, §§4.1–4.4): redundancy, discontinuity, fusion, and form-based form. Agreement is the leading instance of the first. Where one unit of meaning is expressed by several forms, "one of the forms is not necessary; it is redundant, as it does not provide any additional information", and clausal and phrasal agreement are catalogued in that category (Leufkens, 2015, 18, §§4.1.3–4.1.4).

That's the opposite verdict from this paper's on the same configurations. Subject–verb agreement is the paradigm R for which a property remains accessible through a mediator; on the one-to-one conception it's redundant marking, and a language with more of it is less transparent. Both verdicts are correct within their own question. They're different questions: whether a language's levels map one-to-one, and whether a particular relation can reach a property across a mediator that would otherwise conceal it. Call the first MAPPING TRANSPARENCY and the second ACCESS TRANSPARENCY.

English number-transparent QNs cross-classify. They're an access-transparency instance. Being agreement, they're also a mapping-transparency violation. A language could reduce its agreement, becoming more transparent in Leufkens's sense, and thereby lose the very relations that make access transparency visible. The divergence isn't a disagreement to be resolved by defining terms more carefully, because the two programmes individuate by different relata: levels of organization in one, mediator-property-relation triples in the other. An umbrella that delivered opposite verdicts on one construction while both verdicts stood wouldn't be a kind, and that's what §5 concludes on independent grounds.

Two further payoffs. Leufkens's inventory is organized, so it supplies a principled comparison class rather than another loose sense of the word, and the redundancy/discontinuity/fusion/form-based-form taxonomy is a checklist this paper's cases can be run against. And her finding that every language sampled has some non-transparent features, most often of the redundancy type (Leufkens, 2015), means agreement relations are common, so they're where a search for mediated access is most likely to find candidates. That says nothing about how often the search succeeds.

4.5 An objection and a reply

A reader might object that the umbrella schema looks like a universal descriptive category smuggled in as a comparative concept. The sharpest form of the objection targets the non-instance test: *accessible through* presupposes a concealing counterpart, and for the English cases that counterpart is head-driven agreement, which is independently the unmarked pattern, a language-particular fact. If the schema's deepest primitive is parasitic on an English-shaped default, then calling it universally applicable looks like categorial universalism in disguise.

The reply isn't that the schema is "just a question": a universal category can be framed as a question too, so that move wouldn't tell a comparative concept apart from a category in disguise. The reply is that the counterpart the objection points to is fixed *language-internally*, which is how comparative concepts are meant to work. The concealing counterpart is not assumed to be English head-driven agreement; it's whatever same-type variant the language under study attests, by the procedure set out earlier in this section. So the schema doesn't import an English default; it inherits each language's own.

The limitation this concedes is the one §2.3 already states: deciding what counts as the same construction-type with and without concealment is itself a language-internal analysis, so the diagnostic is crisp where that individuation is independently motivated and softer where it isn't. Haspelmath's (2010, p. 665) criterion for a good comparative concept, that it can be more or less well suited to the task of permitting crosslinguistic comparison, is exactly the standard the umbrella schema is meant to meet.

Newmeyer's specific argument deserves a direct response. His case is that typological generalizations must reference not just a universal concept but the specific form realizing it: negative particles and negative auxiliaries express the same concept, but only the latter pattern with verbs in word-order typology, so a generalization stated over "negation" misses the regularity (Newmeyer, 2007, p. 136). The point is correct on its own terms. What follows from it is weaker than crosslinguistic *descriptive* categories, because the formal reference it demands enters at the X slot, and the slot and its values are different things. The slot is *typed* with general-formal notions in Haspelmath's (2010, p. 666) sense: construction, clause, NP, dependency domain. A value of the slot is a language-particular configuration fixed by that language's criteria, the number-transparent QN construction in English and the configurations the analyst identifies in each comparison language. The same holds for P and R: the slots are abstractly typed, while their values are particular number categories, grammatical relations, and agreement systems. Nothing here requires equating configurations token-for-token across languages. The concealing counterpart that gives the non-instance test its grip is itself such a general-formal concept, not an imported universal category.

One limit on the position taken here: Haspelmath's distinction is itself contested, and this paper doesn't adjudicate. It's adopted as the safer default, since it requires no commitment to descriptive-category universality, and a reader more comfortable with categorial universalism can take the schema as a working concept whose status is secondary to its methodological pay-off.[10] One consequence carries into §5: cross-linguistic projection at the profile level isn't entailed by cross-linguistic projection at the comparative-concept level. Asking the question everywhere doesn't make the answers travel.

[10]Croft's (2001) Radical Construction Grammar would route the same commitment through construction-similarity hierarchies rather than comparative concepts; the schema is meant to sit above any single construction-grammar variant, so that route is left open rather than argued against.

5 Projectibility profiles

Section 1 made projectibility a secondary question. A positive access relation may support induction, but the comparative concept itself carries no predictions. This section records which grammatical profiles support further projection and what appears to stabilize them. The primary field is syntax, with targets such as agreement, syntactic-category licensing, and complement selection.

The distinction matters because ACCESS TRANSPARENCY and PROJECTIBILITY PROFILE answer different questions. Applicability asks whether a case fills the schema's roles and controls. A relational verdict asks whether the relation reaches the bearer across the mediator. A profile verdict asks whether an observable base supports projection to further behaviour. The three verdicts should travel separately.

5.1 Applicability and profile verdicts

The umbrella schema is a comparative concept, not a single profile. Its grammatical applications vary in X, P, and R: QNs are the primary demonstration and TFRs a conditional grammatical illustration; Spanish and Tsez are controlled extensions; the Appendix B cases are candidate topologies or boundary cases. No common stabilizer licenses projection across all of them. The schema's contribution is the four-gate applicability protocol and its three-way verdict, not a prediction about every case called *transparent*.

Table 3 assigns the roles and controls case by case. A blank counterpart marks a missing control rather than a negative empirical result. Table 4 then records each case's applicability and relational verdict, in the protocol's terminology (positive, non-instance, or indeterminate), alongside the separate profile verdict.

Table 3: Roles and controls filled, case by case.

Configuration	Property, and its bearer	Mediator	Relation (diagnostic)	Concealing counterpart
QN + *of*-complement	number/count profile, borne by the complement	the QN, morphologically mismatched	agreement (verb form)	head-driven agreement
TFR, parent	number profile, borne by the nucleus	*what*	agreement (verb form)	ordinary fused relative (family-level, weaker)
TFR, attributional daughter	syntactic category, borne by the nucleus	*what*	matrix selection (category flexibility)	evidential daughter, NP nuclei only
Collective NP	member plurality, borne by the head itself	– (same sign)	agreement (verb form)	–
Coordination	summed number, borne by no single conjunct	–	agreement (verb form)	–
Deferred reference	number, borne by the designatum	the subject NP	agreement (verb form)	–

Configuration	Property, and its bearer	Mediator	Relation (diagnostic)	Concealing counterpart
Spanish partitive subject	number, borne by the complement	*mayoría*	agreement (verb form)	singular option, dispreferred
Tsez complement clause	noun class, borne by the embedded absolutive	the clause	matrix agreement (class prefix)	class IV, the clausal default
Aleut (App. B)	person/number, borne by an unpronounced possessor	the possessed NP	anaphoric verb morphology plus subject case	overt-possessor alternant
Lardil/Kayardild (App. B)	clausal TAM, borne by the clause	the dependency domain	dependent-nominal morphology	–
Chamorro (App. B)	grammatical relation of the extractee	the extraction path	path morphology	–
Obviation (App. B)	possessor obviation, borne by the possessor	the possessed NP	predicate tracks the head's derived feature	–

Table 4: Applicability, relational, and profile verdicts.

Case	Applicability and relational verdict	Profile verdict
Number-transparent QNs (§2.1)	Positive: complement number/count reaches agreement through the QN	Strongest candidate; selected agreement and determiner tests support projection, but the full cluster remains open
Transparent free relatives (§2.5)	Analysis-relative: positive under Kim's constructional analysis, nucleus properties reaching agreement and selection through *what*; analysis-neutral: indeterminate	Second syntactic candidate; pilot verb-class evidence, not a completed profile audit
Collective NPs (§2.2)	Non-instance: bearer and mediator are layers of one sign, not distinct elements	None claimed
Ordinary coordination (§2.3)	Non-instance: no mediator conceals the summed features	None
Deferred reference (§2.3)	Non-instance: no bearer occurs within X	None

Case	Applicability and relational verdict	Profile verdict
Spanish partitive subjects (§4.3)	Positive with a weak counterpart: complement number reaches agreement despite a dispreferred singular option	No English-like profile shown
Tsez long-distance agreement (§4.3)	Positive: embedded noun class reaches matrix agreement against a class-IV counterpart	None claimed
Aleut (§B)	Positive with qualification under Merchant's analysis of a null possessor	None claimed
Lardil/Kayardild (§B)	Indeterminate: the inward relation is described, but a same-family counterpart hasn't been established	None claimed
Chamorro (§B)	Indeterminate: path morphology is a candidate relation, without the required counterpart	None claimed
Obviation (§B)	Non-instance or boundary case: the predicate tracks a feature relayed onto the head	None claimed

The table separates a missing control from a failed relation. A case can be positive while its profile remains untested, as with Tsez and the Spanish partitive. A proposed profile can also weaken while the local grammatical relation survives, as the sparse inanimate *bunch* result shows for the QN discussion. Indeterminate cases shouldn't be promoted to positive instances merely because their topology is suggestive.

5.2 Profiles, stabilizers, and levels

The profile verdicts in Table 4 carry the evidential grades set in §2. The QN profile is the strongest candidate: its observable base is independently available, determiner, complement, and agreement behaviours covary by subgroup, and the corpus audit supports selected directions while leaving the end-to-end cluster open, with the *piles* result partial prospective support plus a feasibility failure and the status of *rest* an open fork (§2.4). The TFR profile is a weaker, pilot-level candidate, analysis-relative throughout (§2.5). Two scope exclusions bound the claim: Spanish and Tsez show that the access relation travels, not that the English QN cluster travels with it, and the form–meaning cases of §3 remain boundary applications, with no shared compound, constructional, and processing cluster claimed.

Lexical convention is the proposed stabilizer for the QN profile: speakers acquire item-specific slots such as *a lot of* X and *the rest of* X together with their agreement and complement-selection behaviour, which explains why the class is small and why boundary members and sense splits show graded behaviour, and it doesn't require Boyd-style homeostasis. Constructional inheritance is the proposed stabilizer for the TFR profile, a concrete source of covariance for the joint paper to test rather than an established result. Slater's (2015) stable property cluster (SPC) account asks whether a cluster is stably coinstantiated enough to support the inferential uses assigned to it; the QN profile is the strongest candidate for that diagnostic, the TFR profile a weaker one. Khalidi's (2018) causal-network account asks a different question, about ordered causal stabilizers. The QN class appears conventionally stabilized rather than causally hierarchical, so SPC is the relevant assessment here; no Khalidian result is claimed for the umbrella.

Three levels stay distinct: the comparative concept (access transparency, the cross-linguistic methodological tool defined by the schema of §1); descriptive categories (language-particular grammatical content, such as English number-transparent QNs, Spanish partitive subjects, and Tsez long-distance agreement); and projectibility profiles (observable bases licensing non-trivial projection to a target, stabilized by an identifiable source). Kindhood is downstream: a profile that supports stable projection may earn a local-kind label, a profile with partial support remains a candidate, and a useful descriptive category may remain non-projectible. Applicability comes first and profile evidence second; the distinction preserves the comparative concept while remaining cautious about which language-particular patterns earn further inference.

6 Conclusion

This paper defines access transparency as a comparative concept for configurations in which a relation reaches a property borne inside X across a distinct mediator, while an independently established counterpart shows how the mediator or configuration could instead control the relation. The four-gate protocol requires an independent bearer, a distinct mediator, evidence of tracking, and a constructional counterpart. It returns positive, non-instance, or indeterminate verdicts.

That protocol is the paper's primary contribution. Coordination is a non-instance because plural agreement sums conjunct features without mediation. Deferred reference is a non-instance because its relevant bearer lies outside the subject expression. Collective NPs remain a near case because the proposed bearer and mediator are layers of one sign. These exclusions keep access apart from feature summation, whole-configuration construal, and external reference.

The main grammatical cases are English QNs and TFRs, with the latter conditional on the constructional analysis and pilot evidence. QNs provide the strongest candidate profile: agreement, determiner behaviour, and complement patterns covary within a small conventional inventory. The corpus pilot supports selected agreement directions and a determiner-frame prediction for *piles*, but its asymmetric filtering and sparse subject-position data leave the full projective cluster open. The status of *rest* is an analytical fork that shouldn't decide the broader quantity-QN profile.

TFRs provide a second, conditional grammatical illustration. Under Kim's constructional analysis, nucleus number and category remain accessible through *what*; a pilot verb-class check supports the attributional/evidential contrast. Analysis-neutrally the verdict is indeterminate, and the result remains distributional and provisional until category-discriminating environments, judgments, and rival analyses are handled in the joint paper.

Spanish and Tsez show how the protocol can travel without importing English descriptive categories. The further Appendix B cases mark qualified positives, candidate topologies, and boundary conditions. Form–meaning examples show that R can shift beyond agreement, but they remain boundary applications rather than additional demonstrated projectibility profiles.

Projectibility is downstream. Access transparency identifies a relation; a language-particular profile earns further inference only when an observable base covaries with target behaviour and an identifiable stabilizer supports that covariance. The comparative concept itself predicts nothing. That division of labour preserves the concept's comparative reach while keeping its empirical and metaphysical claims proportionate to the evidence.

A Number-transparent NP empirical apparatus

This appendix collects the COCA pilot data behind the §2 claims about the number-transparent quantificational noun class. The body of the paper carries the qualitative claims; the audit trail lives here. It covers the two cluster behaviours with direct corpus data, the partitive/pseudo-partitive contrast (§A.2) and the override direction (§A.3), plus the *bunch* localization (§A.4) and the negative control (§A.5); the remaining cluster behaviours are *CGEL*-coded predictions, not separately corpus-audited. Query specifications, aggregate counts, KWIC spot-checks, and coding notes are in the supplementary material (see Data availability).

A.1 Methodology

Queries were issued through the local Playwright wrapper at `tools/english-corpora/bin/ecorg.mjs` (May 2026). Each query is a surface string and the counts are raw COCA frequencies, filtered for head-versus-modifier use, idiom contamination, and sense only where noted. Two confounds recur and both inflate counts in the counter-predicted direction: a MODIFIER CONFOUND, where N in X *of* N is a modifier inside a larger NP (*lots of day-care* for *lots of day*), and a PARSE-SHIFT, where the QN sits in a relative clause and the visible verb agrees with a different head (of 13 raw plural hits for *a lot of money are*, 10 are parse-shifted, 1 is irrealis, 1 is contracted *has*, and 1 is a genuine override).

Filtering was applied asymmetrically: counter-predicted cells were checked by hand more thoroughly than predicted ones, because the wrapper's KWIC mode was returning a Playwright error on every scripted call and checks had to be done manually and selectively. The counts below should therefore be read as attestation with a known bias in the paper's favour, not as an estimated grammatical rate. The query specifications, raw returns, and per-cell coding decisions are in the supplementary material (see Data availability).

A.2 Partitive vs. pseudo-partitive *of*-complements

The contrast: PARTITIVE *of*-complements have definite complements (*of the money*, *of those books*); PSEUDO-PARTITIVE *of*-complements have bare or otherwise non-definite complements in the QN frame (*of money*, *of soldiers*, *of some paintings*).

A.2.1 Plural-only forms (*lots, bags, heaps, loads, oodles, stacks, piles*)

Pseudo-partitive (X of N): *lots of people* 3,878; *lots of money* 1,376; *lots of time* 703; *loads of money* 109; *piles of money* 91; *loads of people* 90; *loads of time* 34; *heaps of money* 15; *heaps of people* 7; plus single-digit hits for several other combinations. Total: 6,311. KWIC checks of the single-digit cells show the expected mixture: *lots of day* is modifier use (*day-to-day*, *day hikes*, *day trips*); *heaps of time* and *piles of time* are genuine pseudo-partitive non-count-complement uses; *piles of people* is literal piling rather than the quantificational use. *Piles* is not among *CGEL*'s [57] core forms but patterns with the plural-only class, so it is included in the totals here.

Partitive (X of the N), hand-picked complements: *lots of the time* 7; *lots of the people* 5; *loads of the day* 1, with 0 for *piles* and *heaps*. Total: 13. Those figures are superseded. A wildcard sweep of the same frame (X *of the* *, one query per form, July 2026) returns *lots* 313 tokens across 235 distinct complements; *stacks* 47/42; *loads* 38/33; *bags* 38/35; *heaps* 14/14; *piles* 12/12; *oodles* 3/3. Total: 465 across 374 distinct complements, some thirty-six times the hand-picked count. The partitive frame is attested for every member of the subgroup, and the zeros in the earlier probe were artefacts of testing four complements.

What the sweep does show is a sense split, and it generalizes the *piles* result of §2.5 across the whole subgroup. Wherever a member has a literal homonym, the determined frame recruits it: *stacks* is dominated by library stacks and smokestacks (*the library* 4, *the feed* 2, *the mill* 2, with singletons in *the plant*, *the power*, *the coal*, *the oil*, *the factories*); *heaps* by literal heaps and colliery spoil (*the dead*, *the brush*, *the dross*, *the Rhondda*); *bags* and *loads* by containers and cargo (*the chips*, *the greens*, *the soil*, *the biosolids*, *the shellfish*). Even *lots* carries a pharmaceutical batch sense (*the drug* 3, *the steroid* 2, *the plasma*, *the methylprednisolone*). *Oodles*, the one member with no literal homonym, returns three hits and all three are quantificational (*the cute*, *the sheeple*, *the stuff*). The clearly quantificational partitives concentrate on complements naming an already-established set: *the same* (*lots* 11, *loads* 2), *the stuff* (*lots* 3, *bags* 3, *loads* 3, *heaps* 1, *oodles* 1), *the latter*, *the usual*, *the other*, and definite human plurals (*the women*, *the people*, *the kids*, *the guests*, *the members*, *the children*). Sense assignment here is read off the complement types, not off inspected concordance lines, so it identifies where the literal reading concentrates without measuring its share.

Indefinite-NP complement (X of a N): *lots of a day* 0; *piles of a day* 0; *heaps of a day* 0; *loads of a day* 0; *lots/piles/heaps/loads of an afternoon* 0. Total: 0. That's a null result over eight specific strings, and given what the wildcard did to the definite cells it shouldn't be read as showing the frame unavailable. Testing it properly would take its own sweep.

No partitive share is reported. The wildcard numerator and the hand-picked pseudo-partitive denominator come from different complement inventories, so dividing one by the other would measure the sampling rather than the language. The defensible statement is qualitative: both frames are available across the subgroup, and the bare complement is by a wide margin the characteristic one.

A.2.2 *Plenty*

Pseudo-partitive: *plenty of time* 4,017; *plenty of people* 1,563; *plenty of money* 668; *plenty of day* 4 (likely modifier). Total: 6,252.

Partitive: the two hand-picked complements return *plenty of the time* 1 and *plenty of the people* 1. The wildcard *plenty of the* * returns 169 tokens across 140 distinct complements, led by *the latter* 7, *the same* 6, *the usual* 4, *the things* 4, *the tools* 3, *the real* 3, *the old* 3. *Plenty* has no literal homonym, so the sense confound affecting *stacks, heaps, bags, loads*, and *piles* doesn't arise, and the count stands as quantificational partitive use. The near-absence reported from the two-string probe was an artefact of the two strings. No ratio, for the reason given above.

A.2.3 *Rest, remainder*

Under the semantic criterion adopted in the body, *rest* and *remainder* are not quantity nouns, and that changes what the two frames mean for them. Van Eynde and Kim (2023, p. 271) define a pseudo-partitive as [N1 *of* N2] in which "N1 denotes a quantity or amount of whatever it is that N2 denotes", and treat the bare N2 as "characteristic" of the construction, the property by which it differs from a genuine partitive, "in which N2 is introduced by a definite determiner". For *lot* and the plural-only forms the semantic and syntactic criteria coincide. For *rest*, they may come apart: a competing analysis could treat bare *the rest of society* as pseudo-partitive by treating *rest* as a part or amount expression. *The remainder of time* remains what is left of time, not an amount of time, but the corpus does not by itself settle the lexical classification of *rest*.

Under the semantic analysis, this subgroup has no pseudo-partitive use, and the earlier × in Table 1's pseudo-partitive column was substantively right. The classification of *rest* is therefore left open rather than treated as a settled negative. What was wrong in the earlier draft was its justification: it starred *the rest of time* as ungrammatical. The string is fine, and common.

Bare complement (the X of N). Hand-picked complements returned *the rest of time* 75; *the rest of people* 12; *the rest of day* 7; *the remainder of time* 3; *the remainder of day* 2; *the rest of money* 1, total 100. A wildcard sweep returns far more: *the rest of* [nn*] gives 2,820 tokens across 696 distinct bare complements and *the remainder of* [nn*] 126 across 96. The leading complements are *society* 442, *humanity* 248, *life* 150, *nature* 78, *mankind* 77, *time* 75, *eternity* 73, *world* 63, *creation* 42, *Americans* 41, *government* 31, *baseball* 26, *civilization* 26, *Congress* 21.

Those totals are not a partitive count, and the sweep has to be sorted by what the complement denotes before it says anything. Two classes fall out.

Where the complement denotes a BOUNDED UNIQUE WHOLE, the partitive reading is available and the determiner is unnecessary. Generic collectives are the productive case: *society* 442, *humanity* 248, *nature* 78, *mankind* 77, *creation* 42, *Americans* 41, *government* 31, *baseball* 26, *civilization* 26, *Congress* 21, and *class* 15. Each denotes uniquely without an article and each has proper parts, so a complement subset is well defined: *scientists and the rest of society*.

Where the complement denotes something *unbounded*, it doesn't. Bare *of time* offers nothing to partition, so *the rest of time* is not a partitive with a bare complement; it is a distinct duration expression meaning "all time henceforth", and it is largely fixed (*for the rest of time*, *till the rest of time*). *Eternity* 73 patterns with it. On the genuine partitive reading the determiner is obligatory, so **the remainder of time* stands as a starred datum once the reading is fixed. *Life* 150 is ambiguous between the two classes ("the rest of one's life", duration; "the rest of life on earth", collective) and isn't counted for either.

That resolves why the four-complement probe looked categorical. *Day, time, people*, and *money* contains no bounded generic collective at all, so it sampled only the class where the bare complement fails, and the 100 tokens it returned are duration uses, possessives, and modifier noise rather than partitives. The frame's productive class was outside the sample.

One caveat on the query. Because [nn*] matches a string position, some hits are the first word of a longer nominal (*the rest of day one*, *the rest of tuition/books/housing*), and *the rest of US* is mostly the country rather than the pronoun. The head of the distribution is unaffected.

Determined complement (the X of the N): *the rest of the day* 1,932; *the rest of the time* 830; *the rest of the people* 272; *the rest of the money* 222; *the remainder of the day* 75; *the remainder of the time* 21; *the remainder of the money* 13. Total: 3,365. These are genuine partitives on both criteria.

Indefinite-NP complement: *the remainder of a day* 1.

Determined-complement share *on matched complements*: $3{,}365/(3{,}365+100+1) \approx 97.1\%$ raw. This figure is computable, because numerator and denominator draw on the same four complements, but note what it does and doesn't compare. It is not a partitive-versus-pseudo-partitive ratio: the classification of *rest* remains open, while *remainder* is excluded under the semantic analysis. It measures how often the whole carries a determiner, within one construction, for *day, time, people*, and *money*. For those four the determiner is strongly preferred, and that says nothing about the several hundred bare complements the sweep found.

A.3 Override-direction agreement

Test of the "near-obligatory in central cases" claim. Subject NP plus finite verb agreement, by complement number/count.

Table A.1: Override-direction agreement in the COCA pilot: full returns by complement number/count.

Subject	Predicted	Plural agr.	Singular agr.	% predicted
a lot of people	plural	4,195 (ARE 3,406; WERE 789)	85 (IS 72; WAS 13)	98.0%
a lot of money	singular	1 (ARE, filtered)	90 (IS 61; WAS 29)	98.9%
a number of people	plural	100 (ARE 52; WERE 48)	0 (4 false positives)	100%
lots of people	plural	348 (ARE 288; WERE 60)	0 (6 false positives/non-standard)	100%
lots of money	singular	0 (1 false positive)	24 (IS 19; WAS 5)	100%
plenty of people	plural	79 (ARE 65; WERE 14)	0	100%
plenty of money	singular	0	6 (IS 4; WAS 2)	100%
the rest of the people	plural	19 (ARE 15; WERE 4)	0	100%
the rest of the money	singular	0	19 (IS 10; WAS 9)	100%

The descriptive output is in the predicted direction at 98–100% across all tested QN+complement combinations after filtering. The discriminating cells, where the head's number and the complement's diverge (*a lot of people*, *a number of people*, *lots of money*, *the rest of the people*), carry the evidential weight; the matched cells (*a lot of money*, *lots of people*, *the rest of the money*) confirm the predicted direction but are equally consistent with head-driven agreement. Filtering was applied more thoroughly to counter-direction cells than to predicted-direction cells, so these figures are descriptive attestation, not estimated grammatical rates. *A lot of money* required filtering: of 13 raw plural-agreement hits, 10 were parse-shifted (the surface string *a lot of money are* appearing inside relative clauses modifying a plural-noun head, e.g., *people who earn a lot of money are successful*; *movies that make a lot of money are the biggest help*; *times when we... make a lot of money are gone*), 1 was the irrealis *were* (*if a lot of money were at stake*), and 1 *'s* hit was singular *has* (matches prediction); only 1 hit was a genuine plural override (*a lot of money are being raised*). Filtered count: 1 plural / 90 singular = 98.9% singular.

The other checked counter-direction cells are artefacts. *A number of people is/was* is headed by *condition*, *suggestion*, *ability*, or *counsel*, not by *number*; *lots of people is/was* is headed by expressions such as *murder*, *hauling*, or *disappearance*, with one non-standard existential hit; and *lots of money are* occurs inside a PP modifying plural *men*. The large *a lot of people is/was* cell remains unfiltered, so its 98.0% descriptive row is conservative.

A.4 *Bunch*: animate vs. inanimate aggregates

Testing *CGEL*'s claim (§18.2, p. 503) that *bunch* tilts collective for inanimate aggregates and transparent for groups of people. Animate *bunch*-subjects are consistently plural across *a bunch of people* and *a bunch of kids*, supporting the transparent half. The inanimate half gets no positive support:

inanimate *bunch* is rare in subject position, its typical use being as object; *CGEL*'s own example *A bunch of flowers was presented to the teacher* returns no hits; and the few inanimate-subject tokens take plural rather than the predicted singular. Counts are small enough that this bears only on whether the contrast projects as a corpus regularity, not on the grammatical claim, and the full returns are in the supplementary material.

A.5 Negative control: referential *the number of*

If the override tracked the quantificational construction rather than the noun *number* or N-of-N configurations in general, then referential *the number of*, where *number* is the head rather than a quantifier, should agree head-driven (singular) even with a plural complement. A COCA check (June 2026) bears this out on the same lexeme. *The number of people* occurs with *is/are* at 3/1 and *the number of cases* at 20/1: head-driven singular, with plural only at attraction level. *A number of people* occurs at 0/52 and *a number of cases* at 0/3: categorical plural override. The counts are small, since the bare subject-verb adjacency is rare, but the direction is unambiguous and the contrast sits on one noun: *number* overrides under *a* and not under *the*. The override is therefore a property of the quantificational construction, not of the lexeme or of N-of-N structure as such.

B Accessibility topologies beyond the worked cases

Spanish and Tsez (§4.3) are the cross-linguistic cases the body of the paper works. This appendix sets out four further accessibility topologies the schema maps, each with the applicability or status condition that would weaken or retire it, plus two notes on what the table's entries do and don't establish. None is promoted to a worked projectibility profile; the point is to show the schema's reach and its edges, and to mark which entries lack a concealing control. Every case remains subject to the §2.3 counterpart test: identify X, P, and R, then ask what same-language pattern would make the proposed access disappear.

Table B.1: Accessibility topologies under the umbrella schema.

Topology	X–P–R binding	Applicability condition and status
Outward endpoint access	X = NP or clause mediator; P = complement or embedded feature; R = external agreement. English QNs and transparent free relatives, Spanish partitives, and Tsez long-distance agreement instantiate this topology.	Fails where only the mediator's own head/class feature controls R, or where the construction-type has no concealing variant. This is the paper's core positive pattern.
Null-source subpart access	X = predicate configuration containing a possessed NP with an unpronounced possessor; P = the recoverable possessor's person/number; R = anaphoric verbal morphology plus subject case. Aleut is the key case (Merchant, 2011, p. 195).	Fails if the omitted possessor patterns like an ordinary overt possessor, or if the anaphoric verbal suffix appears without the missing subpart. Positive with a qualification: Merchant analyzes the missing possessor as syntactically present pro.
Inward dependent access	X = clause or nominal-dependency domain; P = clausal TAM/case value; R = morphology on nominal dependents. Lardil and Kayardild show clausal TAM values on dependent nominals (Nordlinger & Sadler, 2004).	Fails if dependent marking is only local nominal morphology, with no covariation with clausal predication. This is an observational topology; no generative analysis is needed for the comparison.
Path-distributed access	X = extraction path; P = grammatical relation/case of the extracted item; R = wh-agreement morphology along that path. Chamorro supplies the comparison (Chung, 1998, 2004).	Fails if the morphology appears only at the gap or endpoint, or if it doesn't vary with the extracted item's role. Positive as a topology, but extraction remains outside this paper's worked evidence.

Topology	X–P–R binding	Applicability condition and status
Feature relay boundary case	X = possessed NP; P = possessor obviation/proximation; R = possessed-noun morphology and verbal obviation agreement. Kutenai and Algonquian obviation are useful near misses (Dryer, 1992).	Fails as transparency where the predicate sees only a feature already relayed onto the possessed head. Boundary case: transformed access, outside the direct-mediated pattern.

The table is a topology map, not a second set of worked cases. Aleut is the qualified positive; the head-marked and obviation entries remain conditional on the analyses and status conditions stated in the table. Lardil, Kayardild, and Chamorro mark candidate applications rather than established instances because an independently motivated same-family concealing control has not been shown.

Data availability

The corpus material behind §2 and Appendix A.3 is the COCA and COHA pilot described there: query specifications, aggregate counts, raw KWIC checks, coding notes, and the pre-registered protocol and results for the *piles* candidate-member test. These files travel with the paper as supplementary material and are versioned in the project repository at https://github.com/BrettRey/Linguistic_transparency. The supplementary README records the manuscript commit and maps reported counts to their queries.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this manuscript, the authors used Claude Opus 4.7 and Opus 5 (1M context), OpenAI Codex 5.5, and GPT-5.6 Sol (pro) for drafting, editing, and review of manuscript text. After using these tools, the authors reviewed and edited the outputs as needed and take full responsibility for the content.

References

Bell, M. J., & Schäfer, M. (2016). Modelling semantic transparency. *Morphology*, *26*(2), 157–199. https://doi.org/10.1007/s11525-016-9286-3

Bhatt, R., & Walkow, M. (2013). Locating agreement in grammar: An argument from agreement in conjunctions. *Natural Language and Linguistic Theory*, *31*(4), 951–1013. https://doi.org/10.1007/s11049-013-9203-y

Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, *61*(1–2), 127–148. https://doi.org/10.1007/BF00385837

Chung, S. (1998). *The design of agreement: Evidence from Chamorro*. University of Chicago Press. https://press.uchicago.edu/ucp/books/book/chicago/D/bo3629289.html

Chung, S. (2004). Restructuring and verb-initial order in Chamorro. *Syntax*, *7*(3), 199–233. https://people.ucsc.edu/~schung/chung_restructuring.pdf

Croft, W. (2001). *Radical construction grammar: Syntactic theory in typological perspective*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198299554.001.0001

da Cunha, Y., & Gibson, E. (2025). Syntactic complexity phenomena are better explained without empty elements mediating long-distance dependencies. *Cognitive Science*, *49*(8), e70088. https://doi.org/10.1111/cogs.70088

Deal, A. R. (2017). External possession and possessor raising. In M. Everaert & H. C. van Riemsdijk (Eds.), *The Wiley Blackwell companion to syntax* (2nd ed.). Wiley Blackwell. https://doi.org/10.1002/9781118358733.wbsyncom047

Dryer, M. S. (1992). A comparison of the obviation systems of Kutenai and Algonquian. In W. Cowan (Ed.), *Papers of the twenty-third Algonquian conference* (pp. 119–163). Carleton University. https://www.acsu.buffalo.edu/~dryer/kut.algon.pdf

Gibson, E. A. F. (2025). *Syntax: A cognitive approach*. MIT Press. Retrieved April 11, 2026, from https://mitpress.mit.edu/9780262553575/syntax/

Goldberg, A. E. (2006). *Constructions at work: The nature of generalization in language*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199268511.001.0001

Goodman, N. (1955). *Fact, fiction, and forecast*. Harvard University Press.

Grosu, A. (2003). A unified theory of standard and transparent free relatives. *Natural Language and Linguistic Theory*, *21*(2), 247–331. https://doi.org/10.1023/A:1023387128941

Haspelmath, M. (2010). Comparative concepts and descriptive categories in crosslinguistic studies. *Language*, *86*(3), 663–687. https://doi.org/10.1353/lan.2010.0021

Hengeveld, K., & Leufkens, S. (2018). Transparent and non-transparent languages. *Folia Linguistica*, *52*(1), 139–175. https://doi.org/10.1515/flin-2018-0003

Heyer, V., & Kornishova, D. (2018). Semantic transparency affects morphological priming ... eventually. *Quarterly Journal of Experimental Psychology*, *71*(5), 1112–1124. https://doi.org/10.1080/17470218.2017.1310915

Hoeksema, J. (2012). On the natural history of negative polarity items. *Linguistic Analysis*, *38*(1–2), 3–33.

Huddleston, R., & Pullum, G. K. (2002). *The Cambridge grammar of the English language*. Cambridge University Press. https://doi.org/10.1017/9781316423530

Keizer, E. (2007). *The English noun phrase: The nature of linguistic categorization*. Cambridge University Press. https://doi.org/10.1017/CBO9780511627699

Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, *195*(4), 1379–1396. https://doi.org/10.1007/s11229-015-0841-y

Kim, J.-B. (2004). Hybrid agreement in English. *Linguistics*, *42*(6), 1105–1128. https://doi.org/10.1515/ling.2004.42.6.1105

Kim, J.-B. (2011). English transparent free relatives: Interactions between the lexicon and constructions. *English Language and Linguistics*, *17*(2), 153–181. https://doi.org/10.17960/ell.2011.17.2.006

Kim, J.-B. (2017). Mixed properties and matching effects in English free relatives: A construction-based perspective. *Linguistic Research*, *34*(3), 361–385. https://doi.org/10.17250/khisli.34.3.201712.006

Kim, J.-B. (2026a). *Form and function mapping in English syntax: A construction grammar perspective*. Routledge. https://doi.org/10.4324/9781003711919

Kim, J.-B. (2026b, April). *Indeterminacy in English grammar: Focusing on free relative clauses* [Unpublished manuscript, 56 pp.].

Kim, J.-B., & Michaelis, L. A. (2020). *Syntactic constructions in English*. Cambridge University Press. https://doi.org/10.1017/9781108632706

Kim, J.-B., & Reynolds, B. (2026, April 8). *Polyfunctionality of* what *in English transparent free relative constructions: A Construction Grammar approach* (Conference presentation) [Wh-elements in Relative and Adverbial Clauses: Perspectives on Polyfunctionality (WRAPP) Workshop, University of Göttingen].

Kim, J.-B., & Sells, P. (2015). English binominal NPs: A construction-based perspective. *Journal of Linguistics*, *51*(1), 41–73. https://doi.org/10.1017/S002222671400005X

Leufkens, S. (2015). *Transparency in language: A typological study*. LOT. https://dare.uva.nl/record/1/439561

Merchant, J. (2011). Aleut case matters. In E. Yuasa, T. Bagchi & K. Beals (Eds.), *Pragmatics and autolexical grammar: In honor of Jerry Sadock* (pp. 193–210). John Benjamins. https://doi.org/10.1075/la.176.12mer

Newmeyer, F. J. (2007). Linguistic typology requires crosslinguistic formal categories. *Linguistic Typology*, *11*(1), 133–157. https://doi.org/10.1515/LINGTY.2007.012

Nordlinger, R., & Sadler, L. (2004). Tense beyond the verb: Encoding clausal tense/aspect/mood on nominal dependents. *Natural Language and Linguistic Theory*, *22*(3), 597–641. https://doi.org/10.1023/B:NALA.0000027720.41506.fe

Polinsky, M., & Potsdam, E. (2001). Long-distance agreement and topic in Tsez. *Natural Language and Linguistic Theory*, *19*(3), 583–646. https://doi.org/10.1023/A:1010757806504

Potsdam, E., & Polinsky, M. (1999). Long-distance agreement in Tsez. In S. Bird, A. Carnie, J. D. Haugen & P. Norquest (Eds.), *Proceedings of the 18th west coast conference on formal linguistics (WCCFL 18)* (pp. 434–447). Cascadilla Press.

Real Academia Española. (n.d.). *La mayoría de los manifestantes, el resto de los alumnos, la mitad de los presentes, etc. + verbo*. Español al día. Retrieved June 16, 2026, from https://www.rae.es/espanol-al-dia/la-mayoria-de-los-manifestantes-el-resto-de-los-alumnos-la-mitad-de-los-presentes

Reynolds, B. (2026a). *The Homeostatic maintenance of English countability: Bidirectional inference and the stability of grammatical clusters* [Preprint, LingBuzz/009537].

Reynolds, B. (2026b). *Words that won't hold still: How linguistic categories work* [Manuscript under review]. https://github.com/BrettRey/hpc-book

Ritchie, S. (2016). Two cases of prominent internal possessor constructions. In D. Arnold, M. Butt, B. Crysmann, T. H. King & S. Müller (Eds.), *Proceedings of the joint 2016 conference on head-driven phrase structure grammar and lexical functional grammar* (pp. 620–640). CSLI Publications. https://doi.org/10.21248/hpsg.2016.32

San Julián Solana, J. (2018a). La concordancia (*ad sensum*) con sustantivos cuantificadores en español. *Verba: Anuario Galego de Filoloxía*, *45*, 67–106. https://revistas.usc.gal/index.php/verba/article/view/3816

San Julián Solana, J. (2018b). La heterogeneidad estructural de las pseudopartitivas en español. *Círculo de Lingüística Aplicada a la Comunicación*, *75*, 261–286. https://doi.org/10.5209/CLAC.61357

Selkirk, E. O. (1977). Some remarks on noun phrase structure. In P. W. Culicover, T. Wasow & A. Akmajian (Eds.), *Formal syntax: Papers from the MSSB-UC Irvine conference on the formal syntax of natural language* (pp. 285–316). Academic Press.

Slater, M. H. (2015). Natural kindness. *British Journal for the Philosophy of Science*, *66*(2), 375–411. https://doi.org/10.1093/bjps/axt033

Stickney, H. (2007). From pseudopartitive to partitive. In A. Belikova, L. Meroni & M. Umeda (Eds.), *Proceedings of the 2nd conference on generative approaches to language acquisition north america (GALANA)* (pp. 406–415). Cascadilla Proceedings Project. https://www.lingref.com/cpp/galana/2/abstract1580.html

Van Eynde, F., & Kim, J.-B. (2023). Pseudo-partitives in English: An HPSG analysis. *English Language & Linguistics*, *27*(2), 271–302. https://doi.org/10.1017/S1360674322000399

van Riemsdijk, H. (2006). Free relatives. In M. Everaert & H. van Riemsdijk (Eds.), *The Blackwell companion to syntax* (pp. 338–382). Blackwell.

Warren, B. (2006). *Referential metonymy*. Kungliga Humanistiska Vetenskapssamfundet i Lund.

Wechsler, S., & Zlatić, L. (2003). *The many faces of agreement: Morphology, syntax, semantics, and discourse factors in Serbo-Croatian agreement*. CSLI Publications. https://web.stanford.edu/group/cslipublications/cslipublications/site/1575864185.shtml

Wilder, C. (1999). Transparent free relatives. In K. N. Shahin, S. Blake & E.-S. Kim (Eds.), *Proceedings of the west coast conference on formal linguistics* (pp. 685–689, Vol. 17). CSLI Publications.